

The background is a deep blue gradient with a subtle pattern of white dots. Overlaid on this are several faint, white circular elements. A large circular scale with degree markings (160, 170, 180, 190, 200, 210, 220, 230, 240, 250, 260) is prominent on the left side. Other circular patterns, some with arrows indicating direction, are scattered across the frame, creating a sense of motion and complexity.

MACHINE LEARNING

HOW COMPUTERS LEARN LIKE YOU DO!

WHAT IS MACHINE LEARNING?

- Machine Learning (ML) is teaching computers to learn from data
- ML helps in making predictions or decisions without being explicitly programmed
- Four main learning paradigms:
 - Supervised Learning
 - Unsupervised Learning
 - Semi-Supervised Learning
 - Reinforcement Learning

SUPERVISED LEARNING (TEACHER'S HELPER)

- "The computer learns from **labeled examples** (like flashcards with answers)."
- **Example:** "If you show it 10 pictures of 🍏 (labeled 'apple') and 🍌 (labeled 'banana'), it learns to guess new fruits!"
- The machine learns from examples **with answers**
- It finds the pattern between inputs and outputs
- Later, it uses that to predict new answers

SUPERVISED LEARNING - REAL WORLD EXAMPLE

- Showing a machine pictures of animals with names:
 - "This is a Dog"
 - "This is a Cat"
- Machine learns to recognize animals

SUPERVISED LEARNING - APPLICATIONS

- Spam email detection
- Predicting house prices
- Face recognition
- Handwriting recognition

UNSUPERVISED LEARNING (THE DETECTIVE)

- "The computer finds **hidden groups** without any labels!"
- **Example:** "Sorting toys into piles (🚗, 🧸, ⚽) by shape/color—no names needed!"

Visual:

- Mixed toys → grouped by type.

UNSUPERVISED LEARNING - REAL WORLD EXAMPLE

- Giving 1000 photos with no names
- Machine groups similar-looking photos together
 - Group A: All dogs
 - Group B: All cats
- **Unsupervised Learning - Applications**
 - Customer segmentation
 - Product recommendations
 - Market analysis
 - Organizing photo libraries

REAL-LIFE UNSUPERVISED LEARNING

- "Where we see it:"
 - Amazon recommending similar toys.
 - Netflix grouping shows you might like.
- **Image:**
 - A "You Might Also Like" suggestion list.

SEMI-SUPERVISED LEARNING (SMART HELPER)

- "Uses a few labels + lots of guesses!"
- **Example:** "Your teacher labels 5 animals (🐱, 🐶), and you guess the rest!"

Visual:

- 2 labeled animals + 10 unlabeled → computer labels them.

REAL-LIFE SEMI-SUPERVISED LEARNING

- "Where we see it:"
 - Google Photos recognizing faces (some tagged, some not).
Image:
 - A photo album with some faces named, some unknown.

SEMI-SUPERVISED LEARNING - APPLICATIONS

- Face recognition (with few labeled photos)
- Language translation
- Detecting rare diseases

REINFORCEMENT LEARNING (GAME PLAYER)

- "Learns by **trial and error** with rewards!"
- **Example:** "A robot dog gets a 🍖 for fetching a ball and learns to do it faster!"

Visual:

- Robot fails → 😞 → tries again → succeeds → 🎉.

REAL-LIFE REINFORCEMENT LEARNING

- "Where we see it:"
 - Self-driving cars (reward for safe driving).
 - AI playing games (Chess ♟️ , Go).
- **Image:**
 - A self-driving car with a "Safe Driving!" badge.

REINFORCEMENT LEARNING - REAL WORLD EXAMPLE

- Teaching a dog:
 - Sit = Treat
 - No sit = No treat
- Machine gets feedback and improves

SUMMARY TABLE – 1

Learning Type	Learns From	Analogy	Goal
Supervised	Labeled data	Textbook with answers	Predict output
Unsupervised	Unlabeled data	Grouping similar books	Find patterns
Semi-Supervised	Few labels + more data	Few solved + many unsolved problems	Learn better with both

SUMMARY TABLE - 2

Type	How It Learns	Kid Example
Supervised	Labeled examples	Flashcards with answers
Unsupervised	Finds hidden groups	Sorting toys blindly
Semi-Supervised	Few labels + many guesses	Homework with hints
Reinforcement	Rewards for good actions	Training a robot dog